

# What is the MDK (Motor Development Kit) ?

Power and control system solutions at your hands

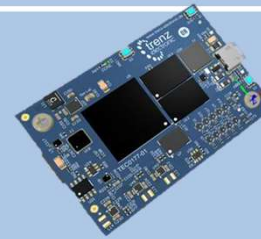
SOLUTION INTEGRATION

## MOTOR DEVELOPMENT KIT

### CONTROL LAYER

#### Universal Controller Board

Xilinx Zynq-7000 SoC - 2x ARM Cor-A9 667MHz + FPGA  
Local supplies  
Flexible and versatile IOs  
Debugging



Avnet Silica  
ON Semiconductor   
Xilinx  
Trenz Electronic

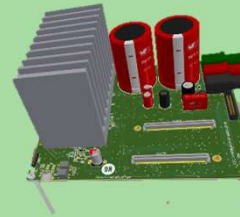
### POWER LAYER

#### Application boards

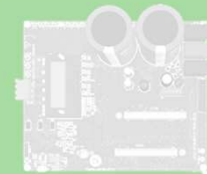
Power stages,  
Aux supply,  
Addl. Interface  
...



Compact IPM 1 kW<sup>1</sup>  
[SECO-1KW-MCTRL-GEVB](#)



SPM31 4 kW  
Power Board



TMPIM  
8-10 kW  
Power board

Power Board  
#4

...

Power Board  
#....

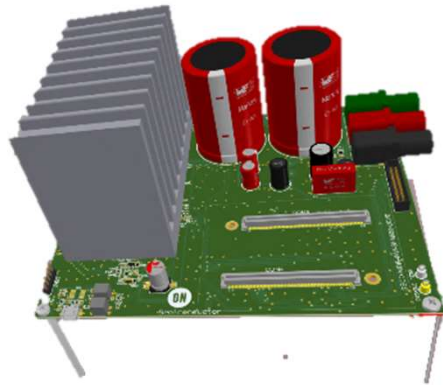
ON

<sup>1</sup> "Arduino Due" Adaption header module required  
Released  
Development

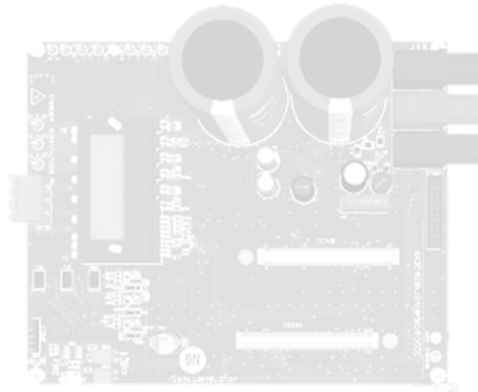
Public Information



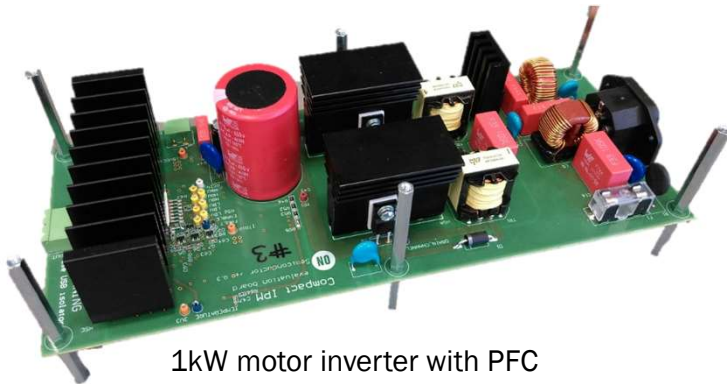
# MDK target applications



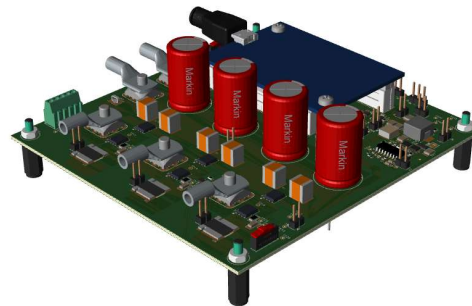
4kW motor inverter board



TMPIM 8 kW 650 V



1kW motor inverter with PFC



48V MOSFET motor inverter

## High-end applications

- 3-phase motor control
- 3-phase PFC
- 3-phase PFC + 3-phase motor
- 3-phase PFC + LLC
- 3-phase / 3 level inverter
- 1-3 Phase Vienna rectifier
- New/experimental topologies

## Targeted motor technology

- BLDC
- BLAC
- Switched reluctance
- PMSM
- Induction
- ...

# Universal Controller Board (UCB) beyond the MDK

UCB's exceptional computational power make it fit for an array of high-end and smart applications

Industrial Power and Motion



Cloud power



Energy Infrastructure



Advanced control strategies

Diagnostics and predictive maintenance

IIoT gateways and protocols

AI, Machine Learning, Deep Learning, ...

Machine vision

...

UCB will be use per default in all new SEC developement boards needing an MCU

# Universal Controller Board Resources

## ➤ MCU

- Zynq 7000 series SoC - Xilinx
- 2x 667 MHz CPU Cortex A9

## ➤ Peripherals

- 10 channel differential ADC (ON Semiconductor NCD9801)
- 12 complementary PWM channels
- Freely configurable digital peripherals
- USB to JTAG/UART interface for configuration/debug
- On-board: ETH PHY and USB OTG
- Off-board: CAN (external transceiver required), USB OTG connector and ETH connector

## ➤ Memory

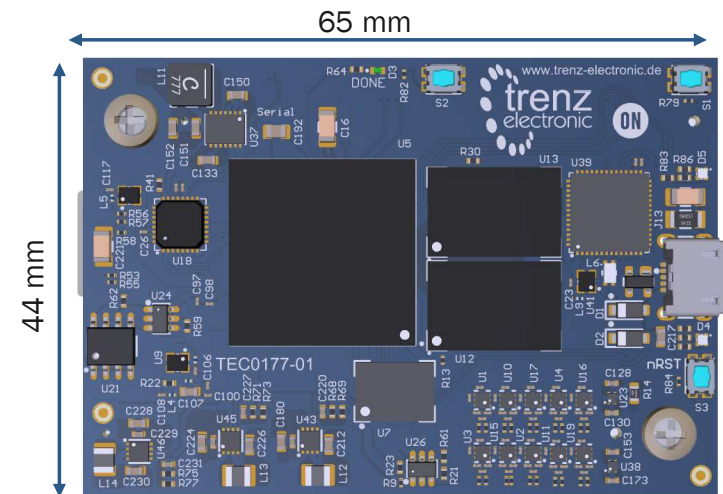
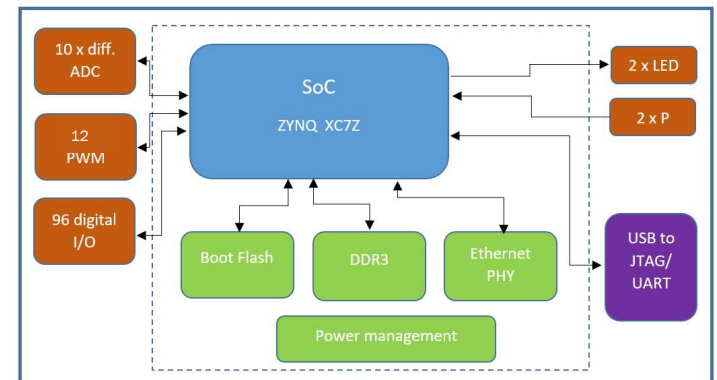
- 512MB RAM
- 512kB EEPROM (ON Semiconductor CAT24C512)
- Micro SD card

## ➤ User interface

- Bootloader capability for user upgradability via UART
- Strata compatibility planned
- LEDs/buttons

Availability Q4

Complete ON Semiconductor solution around Xilinx Zynq SoC



# Module Pin-Out

Power IN

ADC inputs (10x) + AVDD

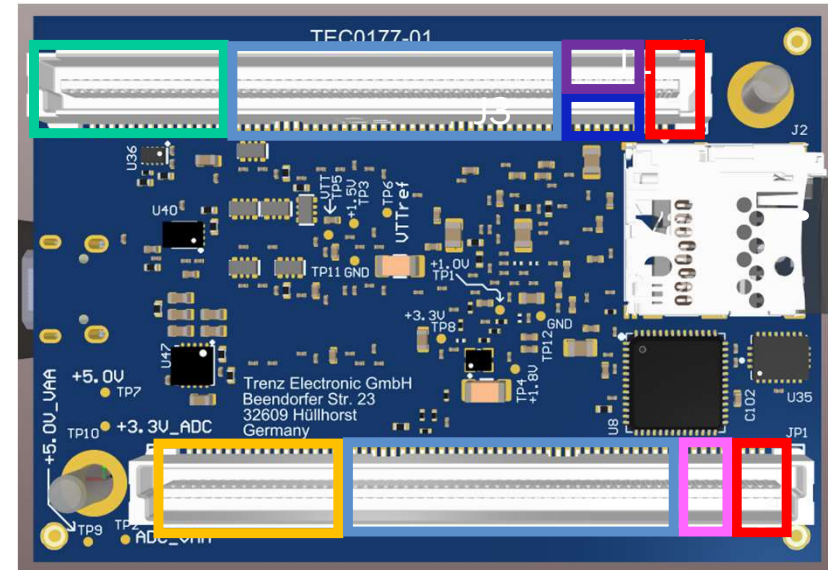
Digital I/O

PWM

Ethernet

USB OTG

JTAG

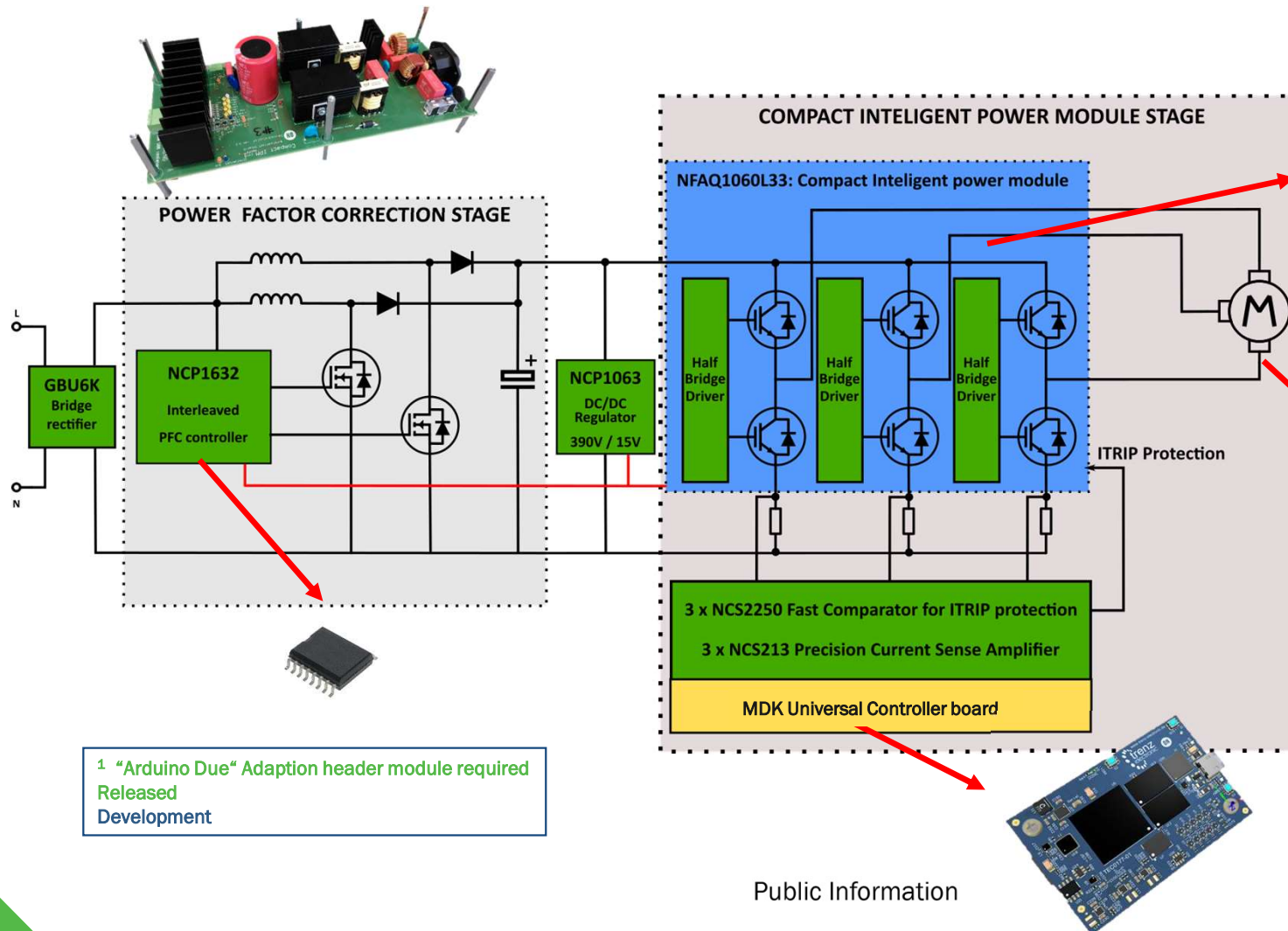


UCB pinout

Concept rendering



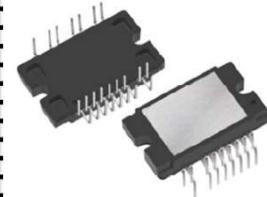
# 1 kW Compact IPM Power Board



Available\*

SECO-1KW-MCTRL-GEVB

\* With Arduino DUE adaptor  
SECO-MDK-DUE-GEVB



1 "Arduino Due" Adaption header module required  
Released  
Development

Public Information



## Q4

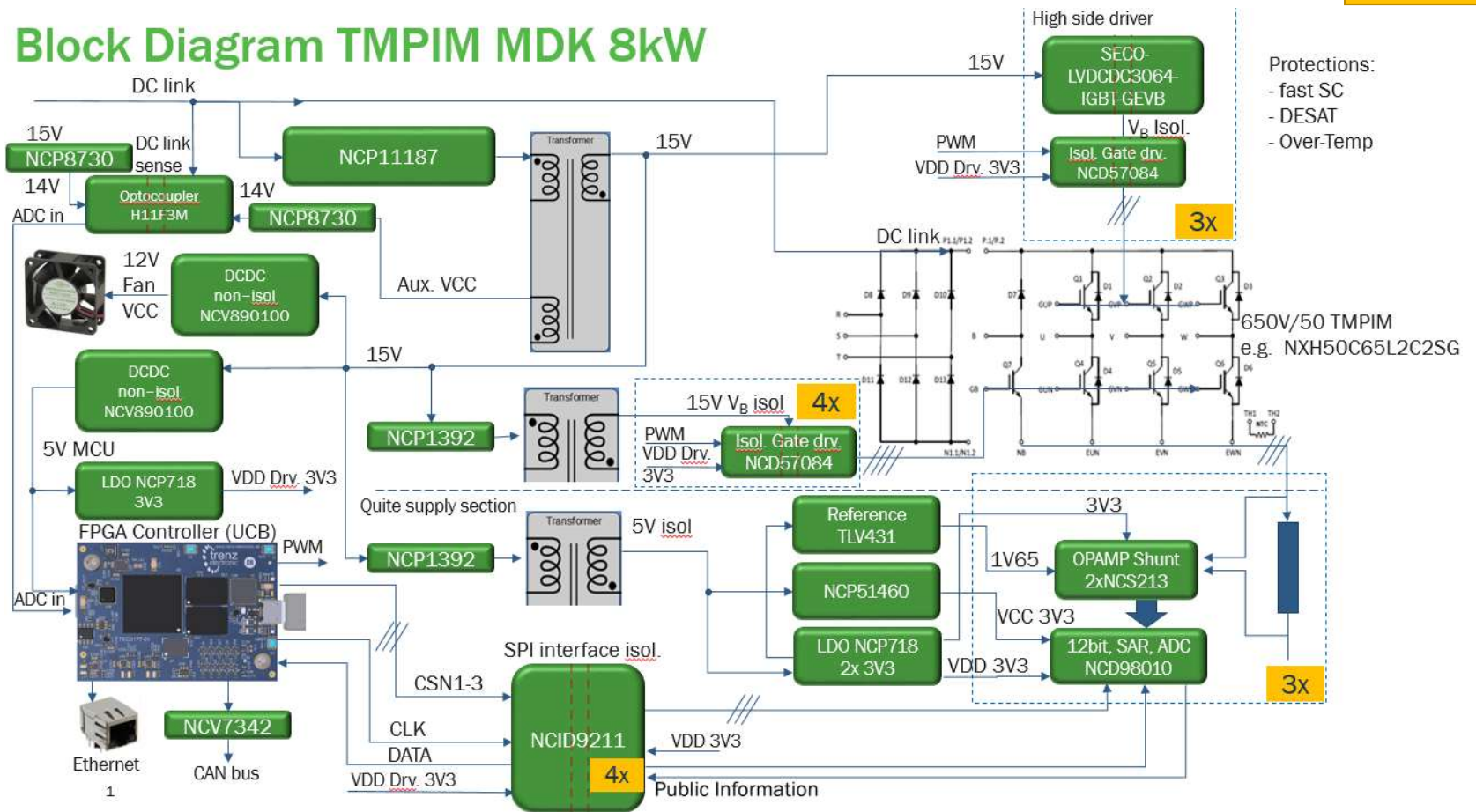


# TMPIM – CBI 8 kW 650 V

Specification file

21Q1

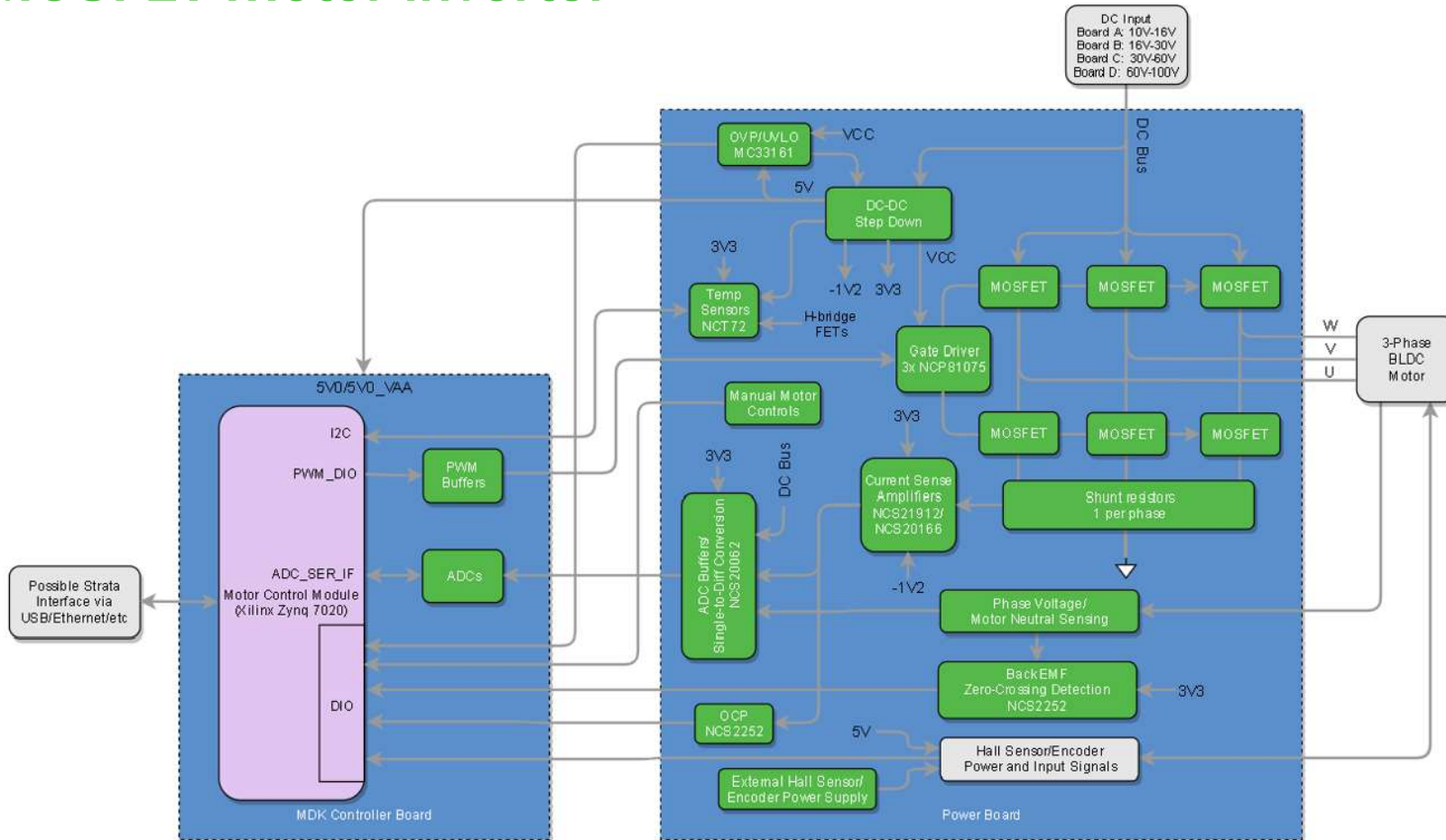
## Block Diagram TMPIM MDK 8kW



Public Information

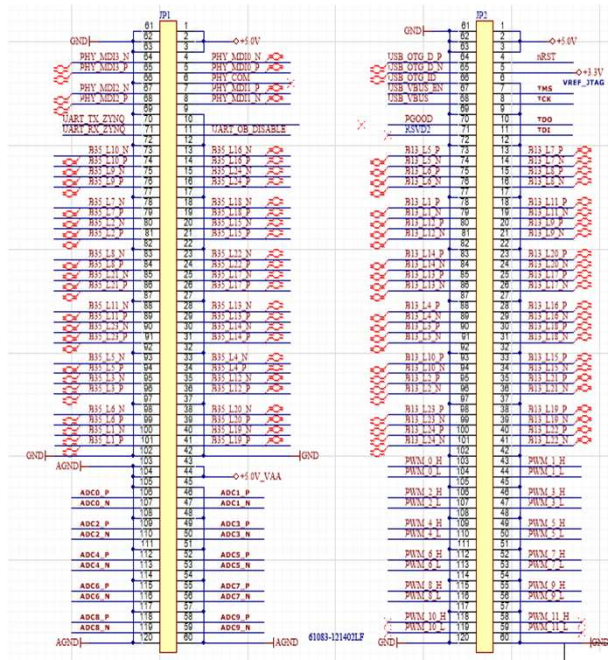


# LV MOSFET motor inverter

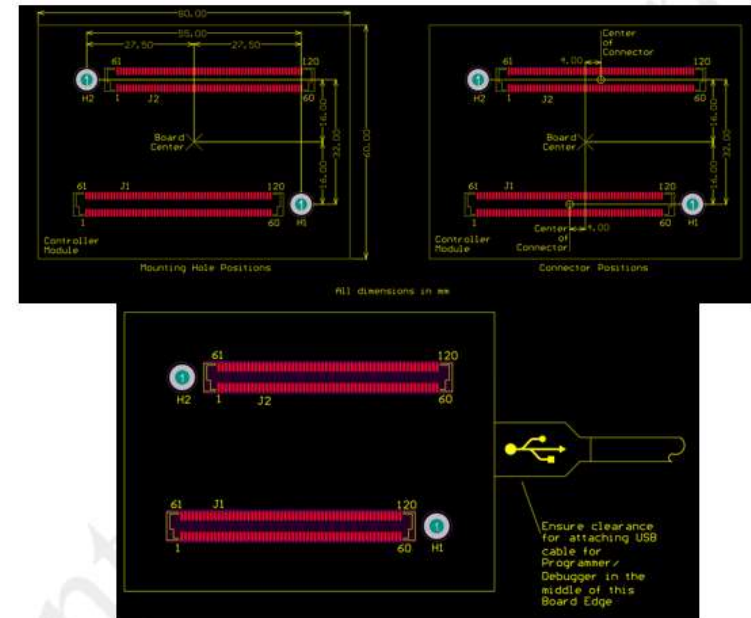


# MDK user guide

- [MDK user guide](#) explains how to integrate the controller board into your custom application.



UCB connector pinout



UCB footprint

Public Information

